

Optimizer-Controlled Implicit Bias and Functional Complexity

Literature review and implications (9.520, Fall 2026)

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October 2, 2026

1 The question and why it is hard

Overparameterized networks have many parameter settings with the same loss; the optimizer picks which one training returns. The project asks whether SGD, Adam and Muon, *at matched loss*, select functions of different complexity, curvature or robustness, and whether swapping optimizers mid-training moves the solution toward the new optimizer’s preferred one. Three obstacles: theory states implicit bias mostly for linear, diagonal or infinitely wide models, as parameter norms that differ by optimizer; experiments report test accuracy, which cannot separate risk-equivalent predictors; and curvature depends on parameterization, so “flatter” need not mean “simpler”.

2 Annotated literature

2.1 Foundations: the optimizer’s geometry selects the solution

Implicit bias belongs to the optimizer and the parameterization together, not to the loss. In linear models a first-order method acts as steepest descent in some norm and converges to the max-margin solution in that norm, but slowly: GD reaches the ℓ_2 margin on separable data only at a logarithmic rate (Soudry et al., 2018), so a finite-time comparison may be far from the limit.

S. Gunasekar et al. (2018a). “Characterizing Implicit Bias in Terms of Optimization Geometry”. In: *ICML*. arXiv: [1802.08246](#)

For mirror descent, natural gradient and steepest descent on underdetermined regression and separable linear classification, asks when the limit depends only on the geometry, not on step size or momentum. For steepest descent on separable data it is the max-margin solution in the chosen norm. Later results for Adam (ℓ_∞) and Muon (spectral norm) follow this template.

S. Gunasekar et al. (2018b). “Implicit Bias of Gradient Descent on Linear Convolutional Networks”. In: *NeurIPS*. arXiv: [1806.00468](#)

GD on full-width linear convolutional networks of depth L converges to a predictor tied to an $\ell_{2/L}$ bridge penalty in the frequency domain; fully connected linear networks reach the hard-margin SVM at any depth. Same optimizer, different function, so any optimizer comparison is also an architecture comparison.

L. Chizat and F. Bach (2020). “Implicit Bias of Gradient Descent for Wide Two-layer Neural Networks Trained with the Logistic Loss”. In: *COLT*. arXiv: [2002.04486](#)

For infinitely wide two-layer homogeneous networks, gradient flow on the logistic loss converges to a max-margin classifier in a non-Hilbertian function space that adapts to hidden low-dimensional structure; training only the last layer gives a kernel SVM without that adaptivity. The bias is stated for *functions*, the level at which we compare optimizers.

A. C. Wilson et al. (2017). “The Marginal Value of Adaptive Gradient Methods in Machine Learning”. In: *NeurIPS*. arXiv: [1705.08292](#)

On constructed separable data, GD and SGD reach zero test error while AdaGrad, RMSProp and Adam approach chance; on deep-learning benchmarks adaptive methods generalize worse even when they train better. The claim we test, that Adam finds different solutions, starts here.

2.2 Measuring functional complexity

We need two independent function metrics. These are defined on the input-output map, so unlike sharpness or weight norms they do not depend on parameterization. Region counts say how many pieces the function has; Jacobian norms

say how steep they are. Both change at constant training loss: regions drift from the training points toward the decision boundary long after interpolation (Humayun et al., 2024), so a matched-loss comparison must also fix training time.

B. Hanin and D. Rolnick (2019). “Complexity of Linear Regions in Deep Networks”. In: *ICML*. arXiv: [1901.09021](#)

At initialization, the number of linear regions of a ReLU network along a line grows linearly in the number of neurons, far below the exponential bound, and stays far below it after training. Region counts are computable and typically small.

R. Novak et al. (2018). “Sensitivity and Generalization in Neural Networks: an Empirical Study”. In: *ICLR*. arXiv: [1802.08760](#)

Across thousands of trained models, the input-output Jacobian norm is small near the training data and correlates with generalization across architectures, optimizers and datasets. It measures slope, complementing region counts.

J. Li et al. (2025). “Understanding Nonlinear Implicit Bias via Region Counts in Input Space”. In: *ICML*. arXiv: [2505.11370](#)

Counts connected input regions with the same predicted label, a parameterization-free measure of implicit bias. Small counts go with simple boundaries and good generalization; larger learning rates and smaller batches give smaller counts, with theory for the learning-rate effect. It compares hyperparameters, not optimizers.

2.3 The GD and SGD side: step size and noise select simpler functions

The best-developed side, and the closest to the course’s training-dynamics line. Step size acts through stability, mini-batch noise through an implicit penalty, scaled by η or η/B . What they buy depends on the model: smoothness in low dimension, a sparse input support, low rank.

D. Qiao et al. (2024). “Stable Minima Cannot Overfit in Univariate ReLU Networks: Generalization by Large Step Sizes”. In: *NeurIPS*. arXiv: [2406.06838](#)

Mulayoff et al. (2021) showed that minima GD can stably reach at step η in univariate two-layer ReLU networks have weighted total variation of f' at most about $1/\eta$. This paper drops interpolation: with label noise σ , stable minima have weighted total variation at most $1/\eta - 1/2 + \tilde{O}(\sigma + \sqrt{\text{MSE}})$, giving a near-optimal $\tilde{O}(n^{-4/5})$ rate. Univariate inputs only.

T. Liang et al. (2025). “Stable Minima of ReLU Neural Networks Suffer from the Curse of Dimensionality: The Neural Shattering Phenomenon”. In: *NeurIPS*. arXiv: [2506.20779](#)

With multivariate inputs, flatness still implies generalization, but the rates worsen exponentially with dimension: flat solutions can “shatter” the data with rarely active, large neurons. Low-norm solutions avoid this, so stability alone cannot explain generalization in high dimension.

S. L. Smith et al. (2021). “On the Origin of Implicit Regularization in Stochastic Gradient Descent”. In: *ICLR*. arXiv: [2101.12176](#)

For small finite η , the mean SGD iterate with random shuffling follows gradient flow on a modified loss that penalizes mini-batch gradient norms, with strength proportional to η/B for small batches. “SGD equals GD plus a step on a regularizer” becomes a theorem, with the sign of the batch-size effect.

P. Beneventano et al. (2024). *How Neural Networks Learn the Support is an Implicit Regularization Effect of SGD*. Preprint. arXiv: [2406.11110](#)

From the instructors’ group. Mini-batch SGD learns the target’s support in the first layer, driving weights on irrelevant inputs to zero through a second-order implicit regularizer proportional to η/B ; full-batch GD needs an explicit penalty to do the same. Relatedly, SGD with weight decay favours low-rank weights (Galanti et al., 2022) and can jump to lower-rank minima but never back (Wang and Jacot, 2024).

2.4 The adaptive side: Adam and Muon

Adaptive methods have their own geometry. Adam behaves like sign descent and Muon like spectral descent, so their limits are max-margin or norm-constrained solutions in ℓ_∞ and in the spectral norm, respectively, not in ℓ_2 . Away from the limit, Adam’s implicit regularizer, unlike SGD’s, can work against small gradient norms.

C. Zhang et al. (2024). “The Implicit Bias of Adam on Separable Data”. In: *NeurIPS*. arXiv: [2406.10650](#)

For logistic regression on linearly separable data, Adam converges in direction to the maximum ℓ_∞ -margin classifier, in polynomial time for a broad class of diminishing learning rates, whereas GD reaches the ℓ_2 margin. Earlier analyses had found that Adam also reaches the ℓ_2 margin, but only because its stability constant ϵ dominates late in training and makes it behave like GD.

- S. Xie and Z. Li (2024). “Implicit Bias of AdamW: ℓ_∞ Norm Constrained Optimization”. In: *ICML*. arXiv: [2404.04454](#)
- Viewing Adam as smoothed sign descent, proves that full-batch AdamW, if it converges, reaches a KKT point of the loss constrained to $\|x\|_\infty \leq 1/\lambda$, where λ is the weight decay. This holds for any differentiable loss, neural networks included, and their experiments confirm it on a small transformer trained on Penn Treebank when $\beta_1 = \beta_2$, though not with the default $\beta_1 = 0.9$, $\beta_2 = 0.999$. So the ℓ_∞ bias holds at any loss, not only in the separable case.
- C. Fan et al. (2025). “Implicit Bias of Spectral Descent and Muon on Multiclass Separable Data”. In: *NeurIPS*. arXiv: [2502.04664](#)
- For multiclass separable data, normalized steepest descent in an entrywise or Schatten p -norm, with or without momentum, converges to the max-margin solution in that norm, with rates. Spectral descent and Muon reach the spectral-norm max margin, and Adam the max-norm margin.
- M. D. Cattaneo et al. (2024). “On the Implicit Bias of Adam”. In: *ICML*. arXiv: [2309.00079](#)
- With finite steps, an optimizer approximately follows gradient flow on a slightly modified loss. For SGD, the modification penalizes large gradients and so pushes training toward flatter regions (Smith et al., 2021). This paper derives the modified loss for RMSProp and Adam: its extra term involves the ℓ_1 norm of the gradient and, for typical hyperparameters, has the opposite sign, slowing the decrease of the gradient instead of penalizing it. Adam therefore need not favour flat solutions.

2.5 Matched-loss comparisons and optimizer swaps

These show the function can change while the error does not. None compares SGD, Adam and Muon at matched loss with two independent function metrics, and the one swap study reports only accuracy. That is the gap.

- F. V. Cortesi et al. (2026). *Same Error, Different Function: The Optimizer as an Implicit Prior in Financial Time Series*. Preprint. arXiv: [2603.02620](#)
- From the course staff. In S&P 500 volatility forecasting, models with identical test loss learn different functions; the optimizer reshapes response profiles and temporal dependence, with about $3\times$ spread in portfolio turnover at similar Sharpe ratio. We extend this matched-loss design beyond one noisy domain.
- B. Vasudeva et al. (2025). “The Rich and the Simple: On the Implicit Bias of Adam and SGD”. in: *NeurIPS*. arXiv: [2505.24022](#)
- For two-layer ReLU networks on Gaussian data, proves GD has a simplicity bias toward a linear boundary with suboptimal margin, while Adam learns richer features and a nonlinear boundary closer to Bayes-optimal; empirically Adam then generalizes better under spurious correlations. Here richness helps, the opposite of the usual lesson from Wilson et al. (2017).
- N. S. Keskar and R. Socher (2017). *Improving Generalization Performance by Switching from Adam to SGD*. Preprint. arXiv: [1712.07628](#)
- SWATS starts with Adam and switches to SGD when a projection-based trigger fires, closing the generalization gap on most CIFAR, Tiny-ImageNet and language-modeling tasks tried. It is the existing swap intervention, but it reports only accuracy, so it cannot say whether the swap changes the learned function.

3 Three key papers

Wilson et al. (2017) (NeurIPS; about 1,200 citations on Semantic Scholar) started the question of whether optimizers differ in what they learn, not just how fast. Its construction proves adaptive and non-adaptive methods can reach opposite test errors on the same data, and its experiments made “SGD generalizes better” a belief later work has had to qualify. Our matched-loss comparisons re-test that claim with modern optimizers.

Gunasekar et al. (2018b) (NeurIPS; about 490 citations; listed in the project description) shows implicit bias depends on parameterization as much as on the optimizer, and characterizes it as a function-space penalty in a nontrivial model, part of the Srebro and Soudry work that made implicit bias a theorem. Design rule for us: hold the architecture fixed, and do not assume results transfer across architectures.

Vasudeva et al. (2025) (NeurIPS 2025) is the most recent top-venue theorem comparing Adam and GD in function space for nonlinear networks. It names a mechanism (GD’s simplicity bias versus Adam’s richer features), predicts boundary shape, and shows “simpler” is not always better. It is the first hypothesis to test on our matched-loss model bank.

4 Implications for theory and practice

1. **Comparing complexity across optimizers is circular unless the metric is neutral.** Each optimizer favours small values of the norm it descends in (Gunasekar et al., 2018a): ℓ_2 for GD, ℓ_∞ for Adam-like methods, spectral for Muon (Xie and Li, 2024; Fan et al., 2025). Measuring complexity in one of these norms favours the matching optimizer by construction. Use metrics defined on the function, such as region counts (Hanin and Rolnick, 2019; Li et al., 2025) and input-Jacobian norms (Novak et al., 2018), and require the two to agree.
2. **Theory: stability explains the GD and SGD side only in low dimension, so structure must do the rest.** Minima stability gives sharp function-space bounds in one dimension (Mulayoff et al., 2021; Qiao et al., 2024), but the rates degrade exponentially with input dimension (Liang et al., 2025). Explanations that survive must use structure: input support (Beneventano et al., 2024), rank (Galanti et al., 2022; Wang and Jacot, 2024), or more generally compositional sparsity, where the relevant dimension is that of each constituent, not the input. Prediction: optimizer differences in complexity should be largest on compositionally sparse targets, where the ℓ_2 , ℓ_∞ and spectral geometries disagree most about which constituents to use.
3. **Practice: matched test loss is not enough to choose an optimizer.** With risk-equivalent predictors, the optimizer still changes robustness, response profiles and downstream decisions (Cortesi et al., 2026; Vasudeva et al., 2025). Report at least one function-level metric next to the loss, especially in noisy domains. A swap at matched loss, as in SWATS (Keskar and Socher, 2017) but scored with function metrics, is a cheap test: if the function moves, the optimizer is a modeling choice, not a tuning detail.
4. **Open question: is the bias set early or applied throughout training?** The break-even picture (Jastrzebski et al., 2020) and the EoS literature suggest hyperparameters fix a curvature regime early. Swapping at different times, scored with two neutral metrics, separates a bias that acts by choosing the basin early from one that keeps acting inside it. This is the causal form of the project’s question, and it connects to the instructors’ work on stability with momentum (Andreyev et al., 2026) and under Muon (Section 2.4).

Other references

- Andreyev, A., A. Ananthkumar, M. Walden, et al. (2026). *Momentum Further Constrains Sharpness at the Edge of Stochastic Stability*. Preprint. arXiv: 2604.14108.
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